

```
[ec2-user@shilpak ~]$ resolvectl status
```

```
Global
```

```
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported  
resolv.conf mode: uplink
```

```
Link 2 (ens5)
```

```
Current Scopes: DNS
```

```
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported  
DNS Servers: 172.31.0.2  
DNS Domain: ec2.internal
```

```
[ec2-user@shilpak ~]$ ip addr show
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
```

```
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
```

```
    inet          /8 scope host lo
```

```
        valid_lft forever preferred_lft forever
```

```
    inet6 ::1/128 scope host noprefixroute
```

```
        valid_lft forever preferred_lft forever
```

```
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
```

```
    link/ether 0a:ff:da:66:68:9f brd ff:ff:ff:ff:ff:ff
```

```
    altname enp0s5
```

```
    altname eni-0cb59b9e0d5ba2038
```

```
    altname device-number-0.0
```

```
    inet          /20 metric 512 brd          scope global dynamic ens5
```

```
        valid_lft 3090sec preferred_lft 3090sec
```

```
    inet6 fe80::8ff:daff:fe66:689f/64 scope link proto kernel_ll
```

```
        valid_lft forever preferred_lft forever
```

```
[ec2-user@shilpak ~]$
```

ec2-user@shilpak:~

[ec2-user@shilpak ~]\$ echo TASK 1 – Linux IP Investigation

TASK 1 – Linux IP Investigation

[ec2-user@shilpak ~]\$ ipconfig

-bash: ipconfig: command not found

[ec2-user@shilpak ~]\$ ifconfig

```
ens5: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
    inet 172.31.17.183 netmask 255.255.240.0 broadcast 172.31.31.255
    inet6 fe80::8ff:daff:fe66:689f prefixlen 64 scopeid 0x20<link>
    ether 0a:ff:da:66:68:9f txqueuelen 1000 (Ethernet)
    RX packets 1115 bytes 2209077 (2.1 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 949 bytes 108138 (105.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 28 bytes 2240 (2.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28 bytes 2240 (2.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

[ec2-user@shilpak ~]\$ hostname -i

fe80::8ff:daff:fe66:689f%ens5 172.31.17.183

[ec2-user@shilpak ~]\$ route -n

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
0.0.0.0	172.31.16.1	0.0.0.0	UG	512	0	0	ens5
172.31.0.2	172.31.16.1	255.255.255.255	UGH	512	0	0	ens5
172.31.16.0	0.0.0.0	255.255.240.0	U	512	0	0	ens5
172.31.16.1	0.0.0.0	255.255.255.255	UH	512	0	0	ens5

[ec2-user@shilpak ~]\$ ip route show default

default via dev ens5 proto dhcp src metric 512

[ec2-user@shilpak ~]\$ resloectl status

-bash: resloectl: command not found

[ec2-user@shilpak ~]\$ resolvectl status

```
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
resolv.conf mode: uplink
```

Link 2 (ens5)

Current Scopes: DNS

```
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    DNS Servers: 172.31.0.2
```

ec2-user@shilpak~

[ec2-user@shilpak ~]\$ echo Task-2 IPV4 Address Analysis

Task-2 IPV4 Address Analysis

[ec2-user@shilpak ~]\$ hostname -i

fe80::8ff:daff:fe66:689f%ens5 172.31.17.183

[ec2-user@shilpak ~]\$ echo "192.168.1.10" | tr '.' '\n'

192

168

1

10

[ec2-user@shilpak ~]\$ ip a

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

inet /8 scope host lo

valid_lft forever preferred_lft forever

inet6 ::1/128 scope host noprefixroute

valid_lft forever preferred_lft forever

2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000

link/ether 0a:ff:da:66:68:9f brd ff:ff:ff:ff:ff:ff

altname enp0s5

altname eni-0cb59b9e0d5ba2038

altname device-number-0.0

inet /20 metric 512 brd scope global dynamic ens5

valid_lft 2817sec preferred_lft 2817sec

inet6 fe80::8ff:daff:fe66:689f/64 scope link proto kernel_ll

valid_lft forever preferred_lft forever

[ec2-user@shilpak ~]\$ curl ifconfig.me

52.72.133.6[ec2-user@shilpak ~]\$ ping -c 3 8.8.8.8

PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.

64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=0.975 ms

64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=0.963 ms

64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=0.986 ms

--- 8.8.8.8 ping statistics ---

3 packets transmitted, 3 received, 0% packet loss, time 2002ms

rtt min/avg/max/mdev = 0.963/0.974/0.986/0.009 ms

[ec2-user@shilpak ~]\$ ^C

[ec2-user@shilpak ~]\$ ping -c 3 1.1.1.1

PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.

64 bytes from 1.1.1.1: icmp_seq=1 ttl=57 time=1.12 ms

64 bytes from 1.1.1.1: icmp_seq=2 ttl=57 time=1.30 ms

^C

--- 1.1.1.1 ping statistics ---

2 packets transmitted, 2 received, 0% packet loss, time 1001ms

rtt min/avg/max/mdev = 1.123/1.209/1.296/0.086 ms

[ec2-user@shilpak ~]\$

```
ec2-user@shilpak:~
```

```
Windows PowerShell
```

```
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```

```
Try the new cross-platform PowerShell https://aka.ms/pscore6
```

```
PS C:\Users\Shilpak> ssh -i .\Downloads\shilpak.pem ec2-user@52.72.133.6
```

```
A newer release of "Amazon Linux" is available.
```

```
Version 2023.12.20260803:
```

```
Version 2023.12.20260817:
```

```
Version 2023.12.20260831:
```

```
Version 2023.12.20260909:
```

```
Run "/usr/bin/dnf check-release-update" for full release and version update info
```

```

      #_
    ~\  #####_      Amazon Linux 2023
  ~~~  \#####\
  ~~~   \###|
  ~~~    \#/_____ https://aws.amazon.com/linux/amazon-linux-2023
  ~~~     V~' '->
    ~~~
      ~~-._. _/
        _/_/_/
          _/m/'
```

```
Last login: Fri Sep 11 10:37:15 2026 from 116.75.172.237
```

```
[ec2-user@shilpak ~]$ hostname -I
```

```
172.31.17.183
```

```
[ec2-user@shilpak ~]$ curl ifconfig.me
```

```
52.72.133.6[ec2-user@shilpak ~]$
```

☐	practuce	i-023842270c8c00255	 Running  	t3.micro	 5/5 checks passed	us-east-1c	ec2-54-175-98-204.com...	54.175.98.204	-
☐	AMI	i-05303d430f1b7e543	 Stopped  	t3.micro	-	us-east-1c	-	-	-
☒	task4	i-02acf77f9d7b70d56	 Running  	t3.micro	 Initializing	us-east-1c	ec2-54-175-98-204.co...	54.175.98.204	-
☐	ubuntuclass	i-06a21616f5b22cee9	 Stopped  	t3.micro	-	us-east-1a	-	-	-

```
ec2-user@shilpak~  
[ec2-user@shilpak ~]$ ping -c 4 google.com  
PING google.com (172.253.63.139) 56(84) bytes of data.  
64 bytes from bi-in-f139.1e100.net (172.253.63.139): icmp_seq=1 ttl=106 time=1.25 ms  
64 bytes from bi-in-f139.1e100.net (172.253.63.139): icmp_seq=2 ttl=106 time=1.24 ms  
64 bytes from bi-in-f139.1e100.net (172.253.63.139): icmp_seq=3 ttl=106 time=1.26 ms  
^C  
--- google.com ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2002ms  
rtt min/avg/max/mdev = 1.238/1.250/1.264/0.010 ms  
[ec2-user@shilpak ~]$ traceroute google.com  
traceroute to google.com (172.253.63.113), 30 hops max, 60 byte packets  
 1  ec2-3-236-61-207.compute-1.amazonaws.com (3.236.61.207)  6.042 ms 244.5.0.95 (244.5.0.95)  4.332 ms ec2-3-236-61-243.compute-1.amazonaws.com (3.236.61.243)  6.021 ms  
 2  240.0.52.131 (240.0.52.131)  0.287 ms 240.4.88.37 (240.4.88.37)  0.311 ms 240.3.16.129 (240.3.16.129)  0.344 ms  
 3  242.0.202.213 (242.0.202.213)  2.965 ms 242.0.202.211 (242.0.202.211)  0.976 ms 242.0.203.215 (242.0.203.215)  0.554 ms  
 4  * * *  
 5  * * 72.14.203.158 (72.14.203.158)  0.585 ms  
 6  * * *  
 7  142.251.70.112 (142.251.70.112)  1.512 ms 142.251.53.0 (142.251.53.0)  1.155 ms 142.251.52.64 (142.251.52.64)  1.632 ms  
 8  192.178.243.4 (192.178.243.4)  1.652 ms 192.178.248.40 (192.178.248.40)  0.624 ms 192.178.110.210 (192.178.110.210)  1.390 ms  
 9  72.14.233.183 (72.14.233.183)  1.351 ms 142.251.49.155 (142.251.49.155)  1.173 ms 142.251.49.19 (142.251.49.19)  1.557 ms  
10  142.250.59.233 (142.250.59.233)  3.174 ms 142.251.245.83 (142.251.245.83)  3.584 ms 142.250.215.189 (142.250.215.189)  1.556 ms  
11  192.178.123.237 (192.178.123.237)  2.834 ms 172.253.71.204 (172.253.71.204)  0.969 ms 72.14.235.116 (72.14.235.116)  2.471 ms  
12  172.253.72.29 (172.253.72.29)  3.206 ms 172.253.72.41 (172.253.72.41)  2.068 ms 172.253.72.37 (172.253.72.37)  2.195 ms  
13  * * *  
14  * * *  
15  * * *  
16  * * *  
17  * * *  
18  * * *  
19  * * *  
20  * * *  
21  * * *  
22  * bi-in-f113.1e100.net (172.253.63.113)  2.018 ms *  
[ec2-user@shilpak ~]$ sudo systemctl status nginx  
● nginx.service - The nginx HTTP and reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)  
   Drop-In: /usr/lib/systemd/system/nginx.service.d  
            └─php-fpm.conf  
   Active: active (running) since Thu 2026-09-10 15:43:16 UTC; 19h ago  
   Process: 1896 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)  
   Process: 1901 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)  
   Process: 1920 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)  
 Main PID: 1926 (nginx)  
    Tasks: 3 (limit: 1059)  
   Memory: 7.2M
```

ec2-user@shilpak~

```
Process: 1920 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
Main PID: 1926 (nginx)
Tasks: 3 (limit: 1059)
Memory: 7.2M
CPU: 112ms
CGroup: /system.slice/nginx.service
└─1926 "nginx: master process /usr/sbin/nginx"
└─1927 "nginx: worker process"
└─1928 "nginx: worker process"
```

```
Sep 10 15:43:16 shilpak systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Sep 10 15:43:16 shilpak nginx[1901]: nginx: [warn] conflicting server name "_" on 0.0.0.0:80, ignored
Sep 10 15:43:16 shilpak nginx[1901]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Sep 10 15:43:16 shilpak nginx[1901]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Sep 10 15:43:16 shilpak nginx[1920]: nginx: [warn] conflicting server name "_" on 0.0.0.0:80, ignored
Sep 10 15:43:16 shilpak systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
```

[ec2-user@shilpak ~]\$ sudo netstat -tulnp

Active Internet connections (only servers)

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State	PID/Program name
tcp	0	0	0.0.0.0:22	0.0.0.0:*	LISTEN	1899/sshd: /usr/sbi
tcp	0	0	0.0.0.0:80	0.0.0.0:*	LISTEN	1926/nginx: master
tcp6	0	0	:::22	:::*	LISTEN	1899/sshd: /usr/sbi
tcp6	0	0	:::80	:::*	LISTEN	1926/nginx: master
udp	0	0	127.0.0.1:323	0.0.0.0:*		1933/chronyd
udp	0	0	172.31.17.183:68	0.0.0.0:*		1710/systemd-networ
udp6	0	0	:::1:323	:::*		1933/chronyd
udp6	0	0	fe80::8ff:daff:fe66:546	:::*		1710/systemd-networ

[ec2-user@shilpak ~]\$ sudo iptables -L -n -v

sudo: iptables: command not found

[ec2-user@shilpak ~]\$ █

ec2-user@shilpak:~

[ec2-user@shilpak ~]\$ echo TASK 5 – Cloud Network Troubleshooting

TASK 5 – Cloud Network Troubleshooting

[ec2-user@shilpak ~]\$ ip addr show

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:da:66:68:9f brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-0cb59b9e0d5ba2038
    altname device-number-0.0
    inet 172.31.16.1/20 metric 512 brd 172.31.16.255 scope global dynamic ens5
        valid_lft 2181sec preferred_lft 2181sec
    inet6 fe80::dff:daff:fe00::1/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

[ec2-user@shilpak ~]\$ ip link show

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 0a:ff:da:66:68:9f brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-0cb59b9e0d5ba2038
    altname device-number-0.0
```

[ec2-user@shilpak ~]\$ route -n

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
0.0.0.0	172.31.16.1	0.0.0.0	UG	512	0	0	ens5
172.31.0.2	172.31.16.1	255.255.255.255	UGH	512	0	0	ens5
172.31.16.0	0.0.0.0	255.255.240.0	U	512	0	0	ens5
172.31.16.1	0.0.0.0	255.255.255.255	UH	512	0	0	ens5

[ec2-user@shilpak ~]\$ ping -c google.com

ping: invalid argument: 'google.com'

[ec2-user@shilpak ~]\$ ping -c 4 google.com

PING google.com (172.253.63.139) 56(84) bytes of data.

64 bytes from bi-in-f139.1e100.net (172.253.63.139): icmp_seq=1 ttl=106 time=1.25 ms

64 bytes from bi-in-f139.1e100.net (172.253.63.139): icmp_seq=2 ttl=106 time=1.24 ms

64 bytes from bi-in-f139.1e100.net (172.253.63.139): icmp_seq=3 ttl=106 time=1.26 ms

^C

--- google.com ping statistics ---

3 packets transmitted, 3 received, 0% packet loss, time 2002ms

rtt min/avg/max/mdev = 1.238/1.250/1.264/0.010 ms

[ec2-user@shilpak ~]\$ traceroute google.com